HW6

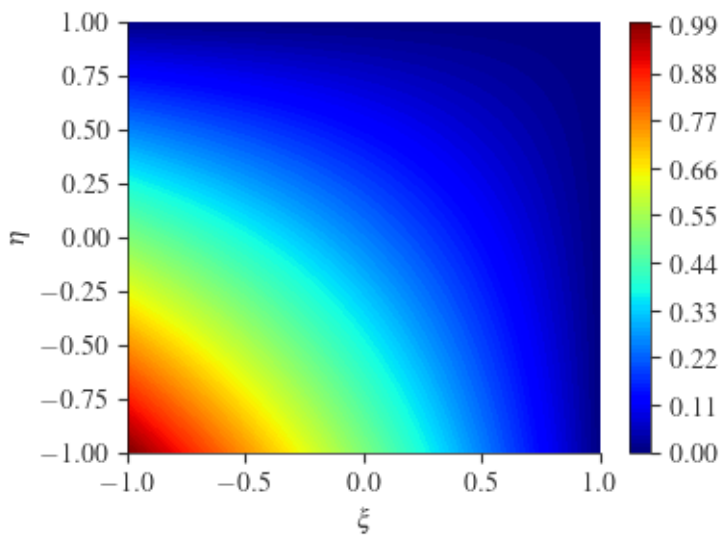
September 25, 2024

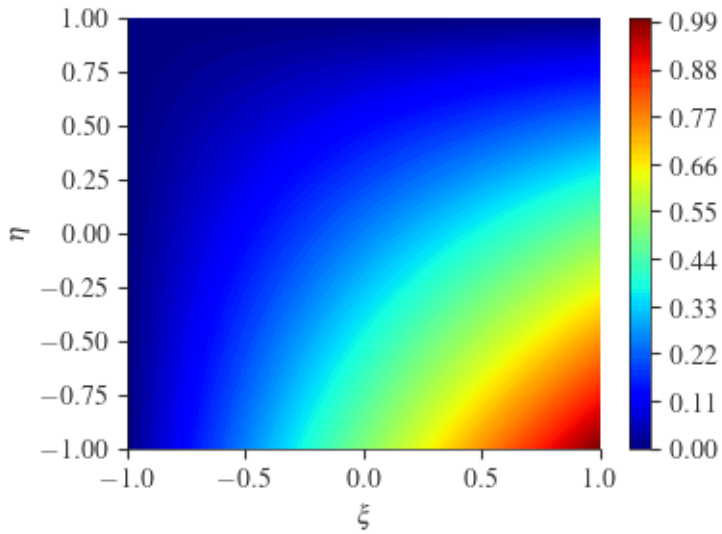
1 Homework 6

1.1 Problem 1

1.1.1 Polynomials of degree [1,1]

[16]:
```
import CE_ME_507_MultiDimensionalBasisFunctions as mdf
import math
degs = [1,1]
nbfs = math.prod([p+1 for p in degs])
for i in range (nbfs):
    mdf.PlotTwoDimensionalParentBasisFunction(i,degs)
```

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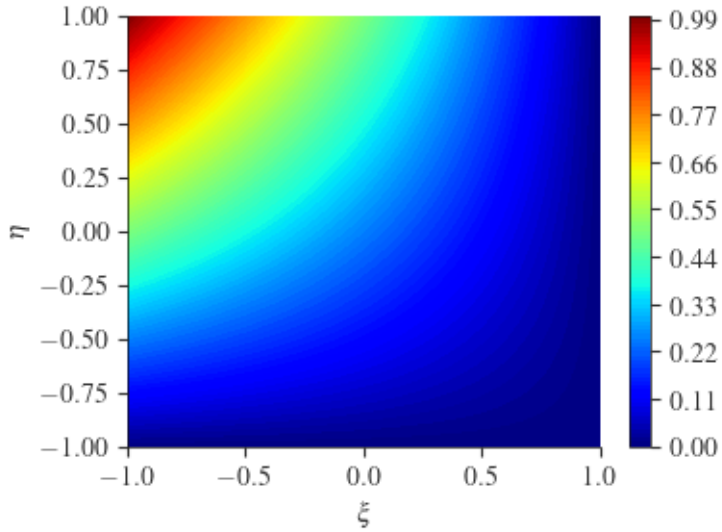
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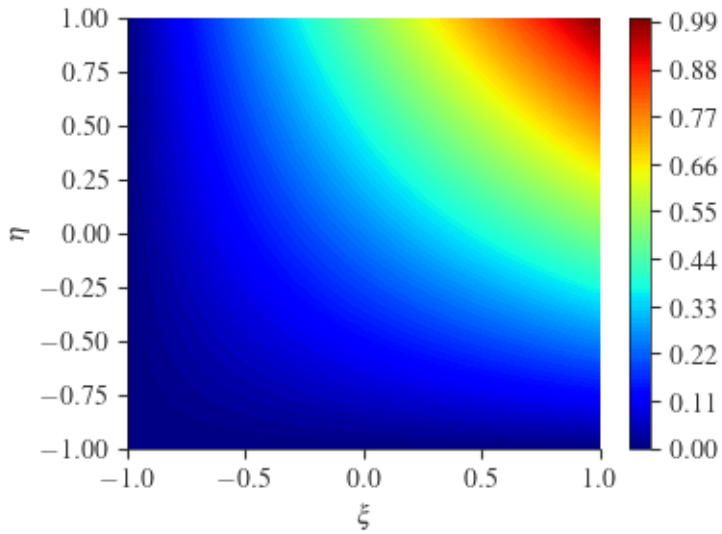

1.1.2 Polynomials of degree [2,1]

[17]:
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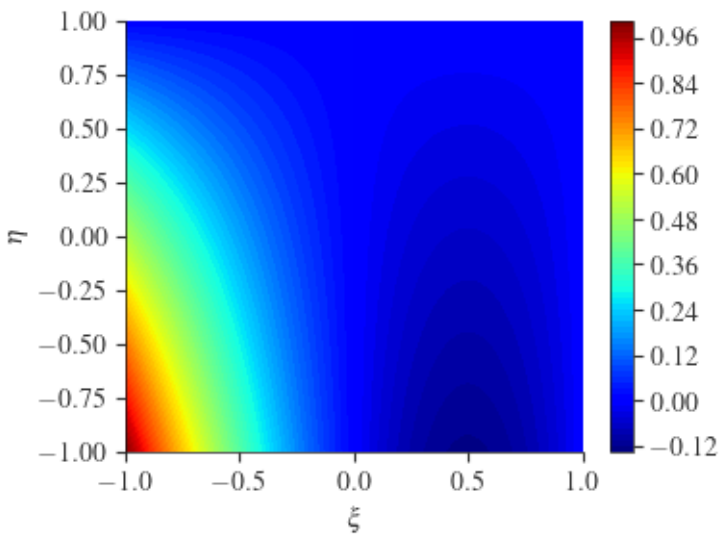
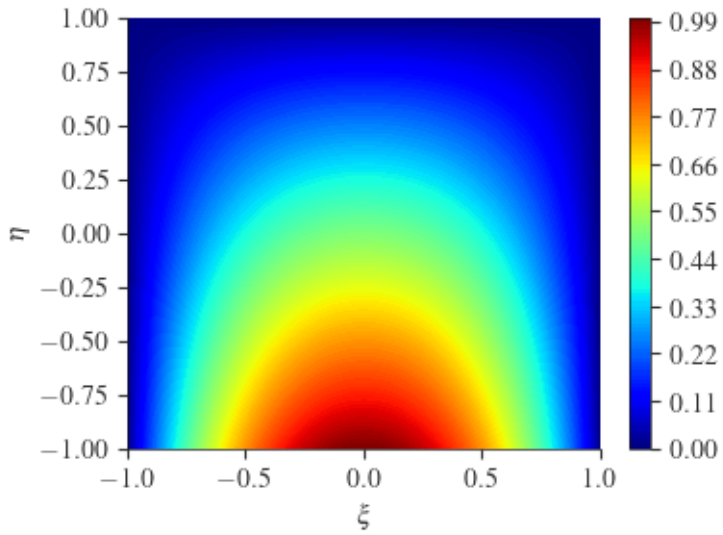
degs = [2,1]
nbfs = math.prod([p+1 for p in degs])
for i in range (nbfs):
    mdf.PlotTwoDimensionalParentBasisFunction(i,degs)
```

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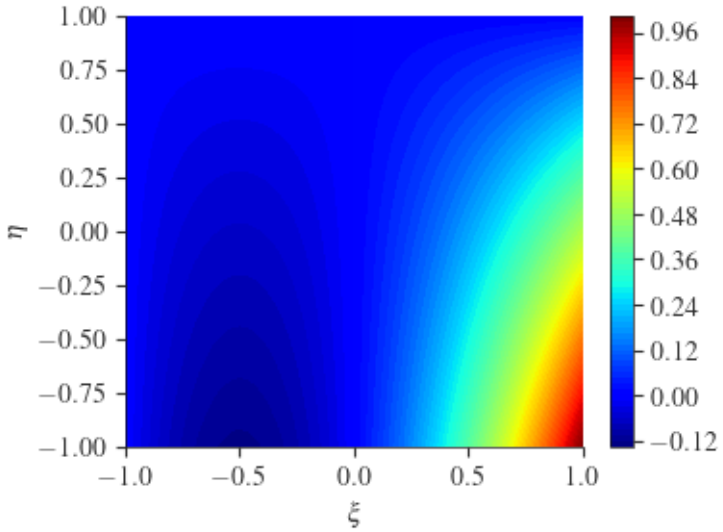
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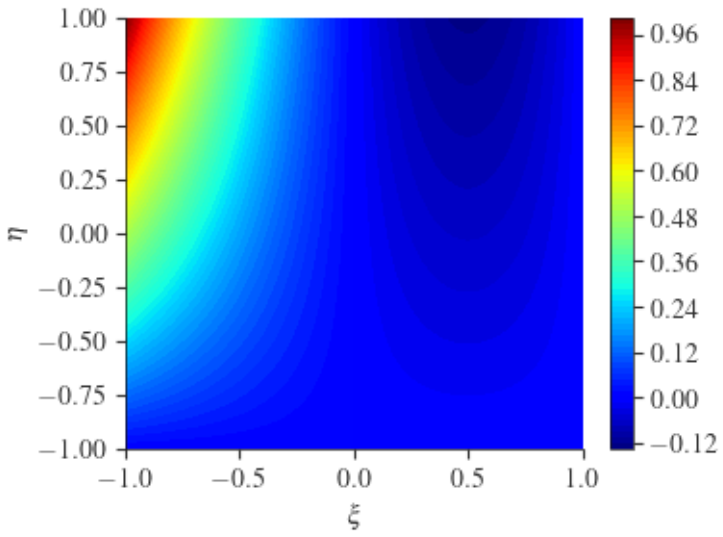
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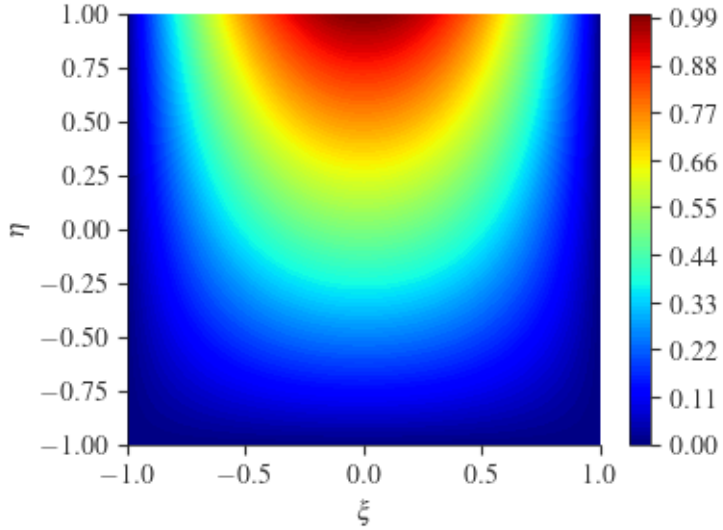
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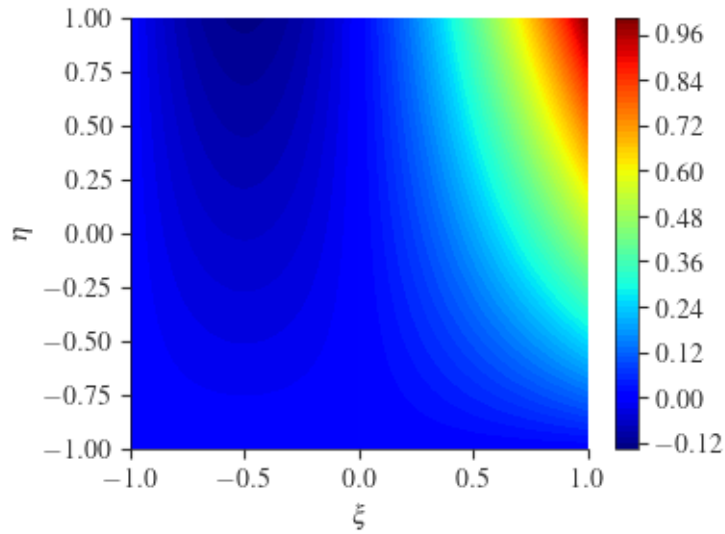

1.1.3 Polynomials of degree [2,2]

[18]:
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degs = [2,2]
nbfs = math.prod([p+1 for p in degs])
for i in range (nbfs):
    mdf.PlotTwoDimensionalParentBasisFunction(i,degs)
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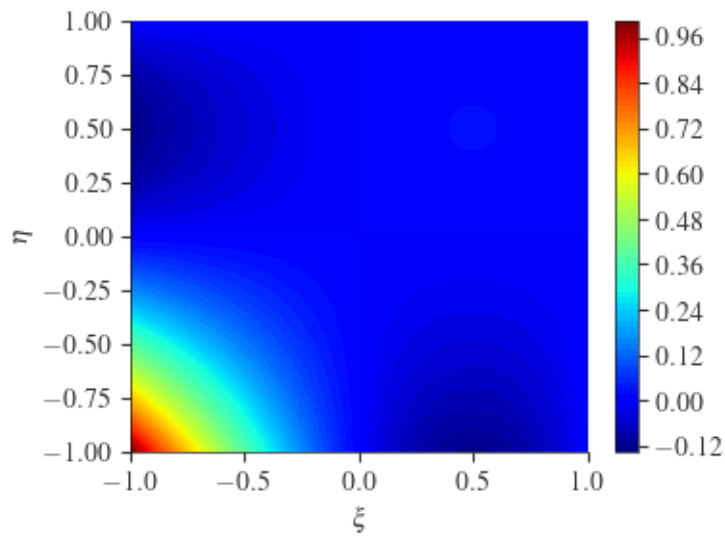

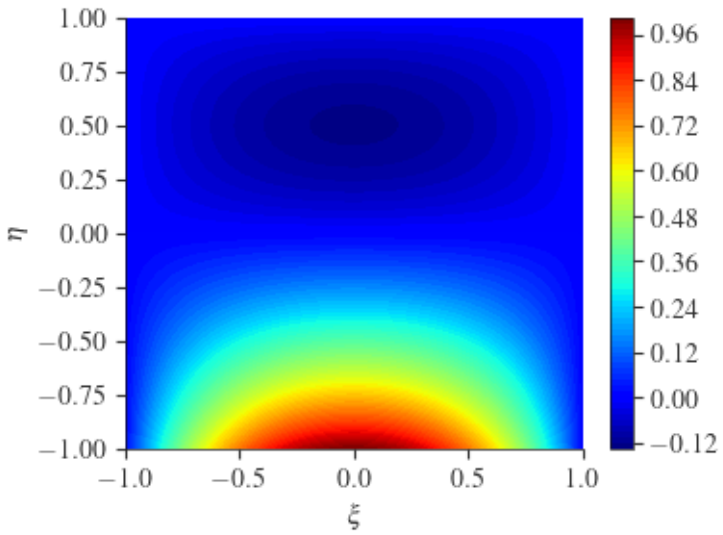
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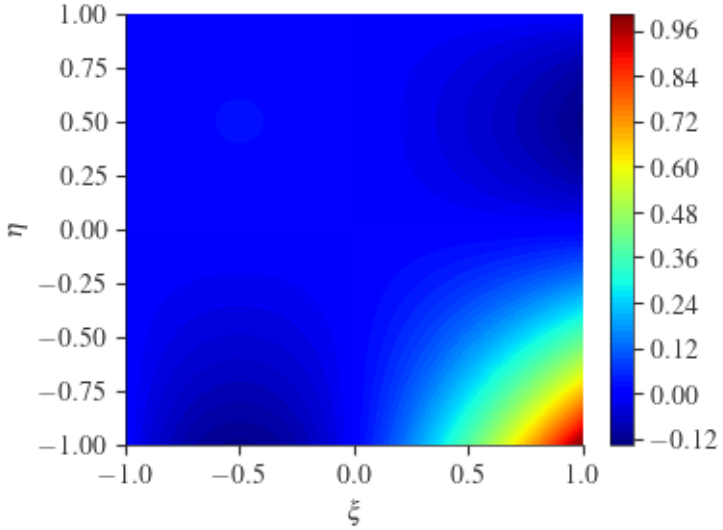
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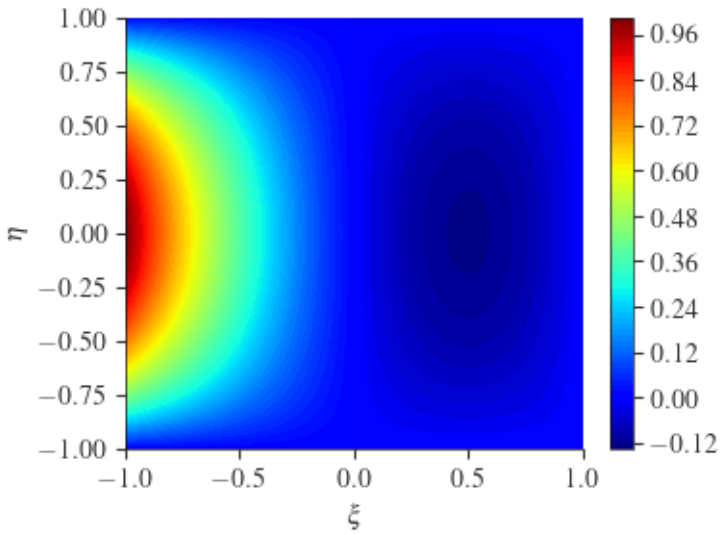
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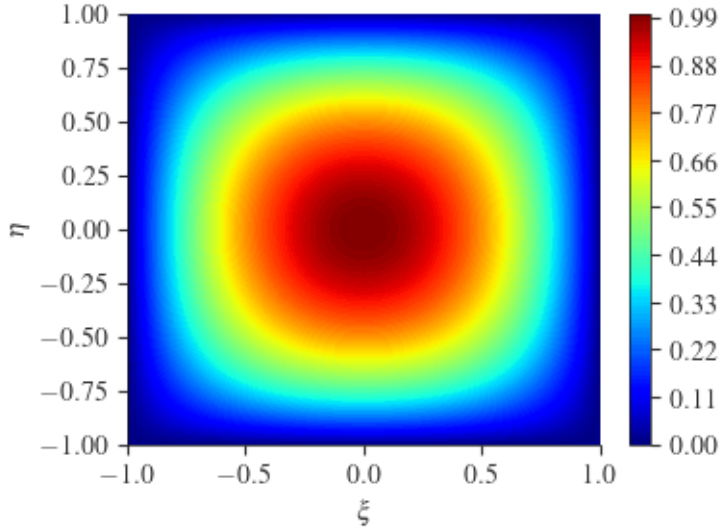
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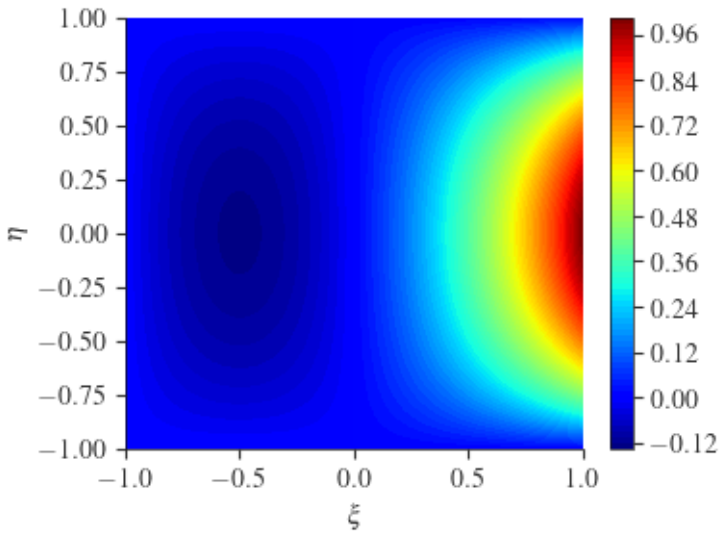
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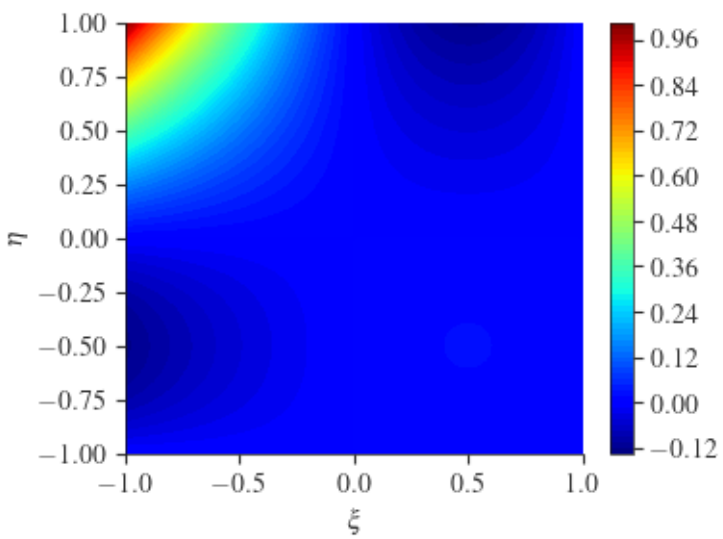
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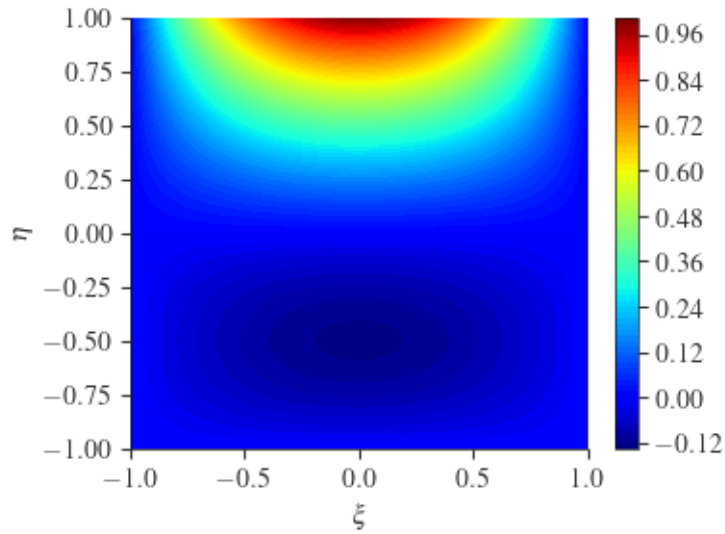
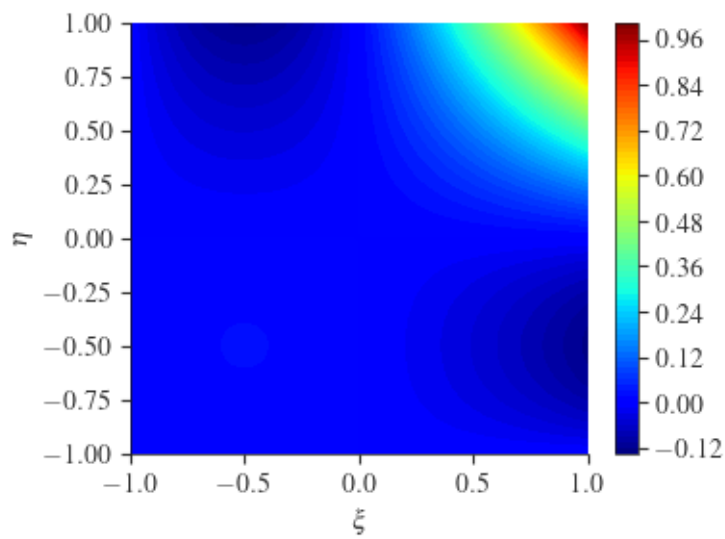
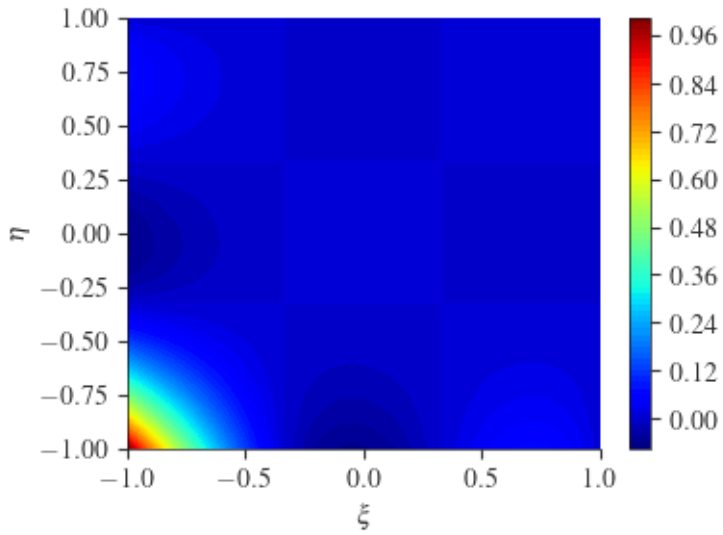




1.1.4 Polynomials of degree [3,3]

[19]:
```
degs = [3,3]
nbfs = math.prod([p+1 for p in degs])
for i in range (nbfs):
    mdf.PlotTwoDimensionalParentBasisFunction(i,degs)
```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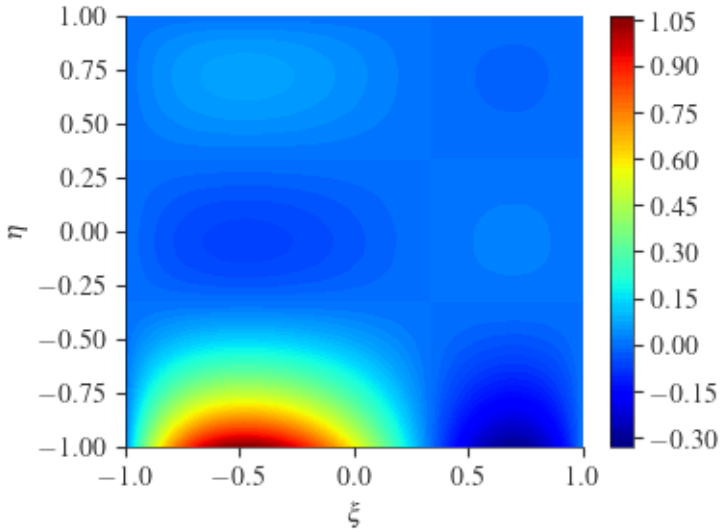
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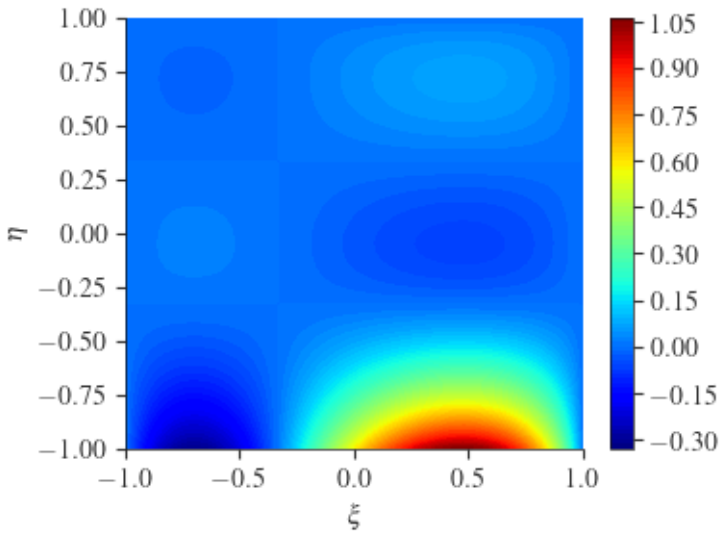
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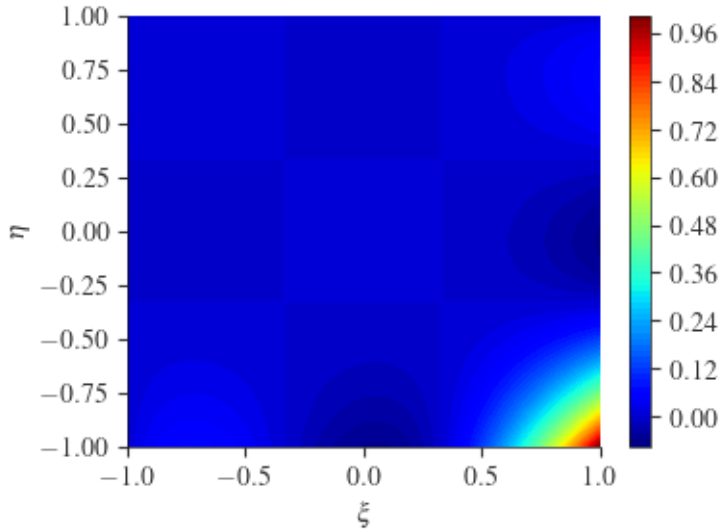
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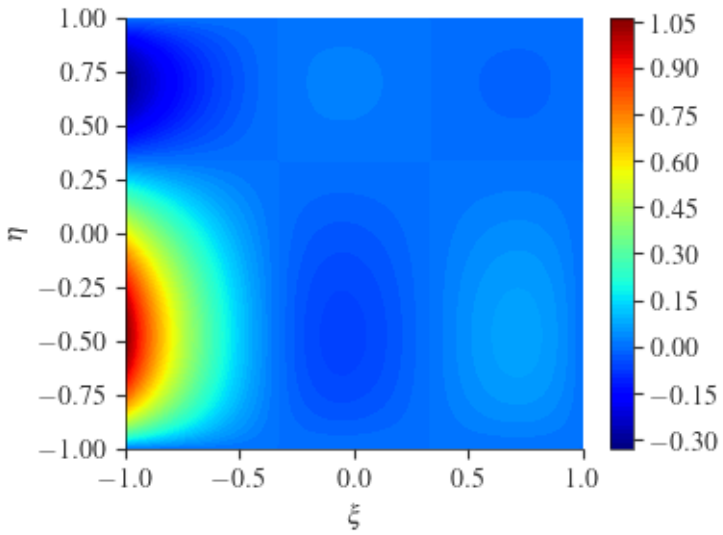
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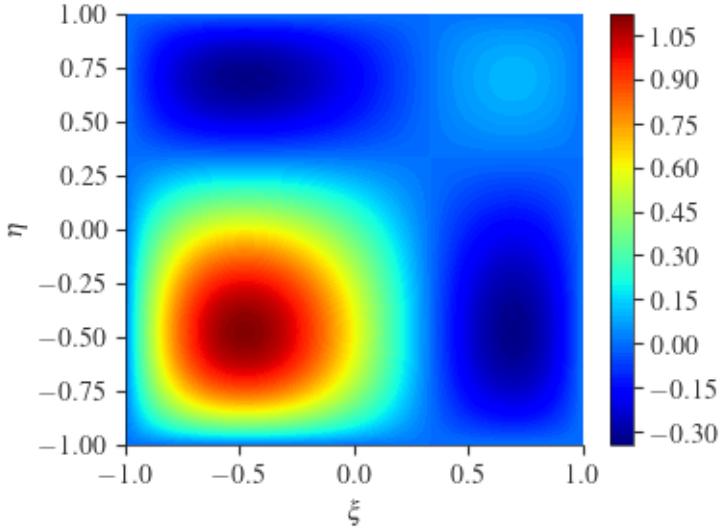
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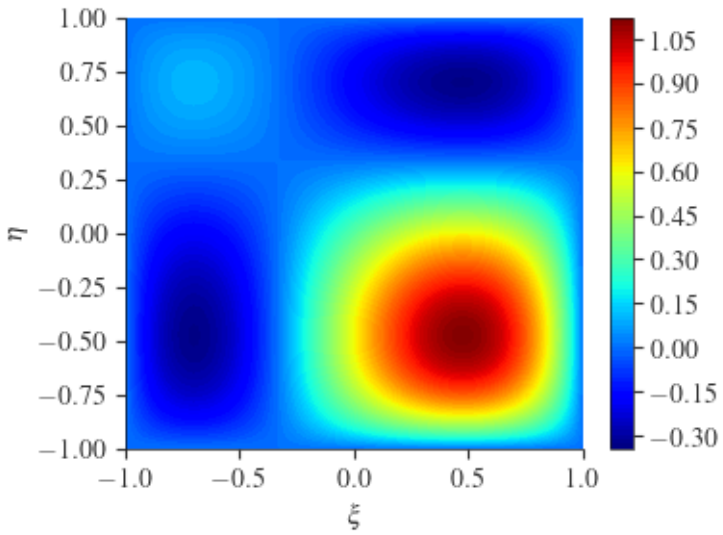
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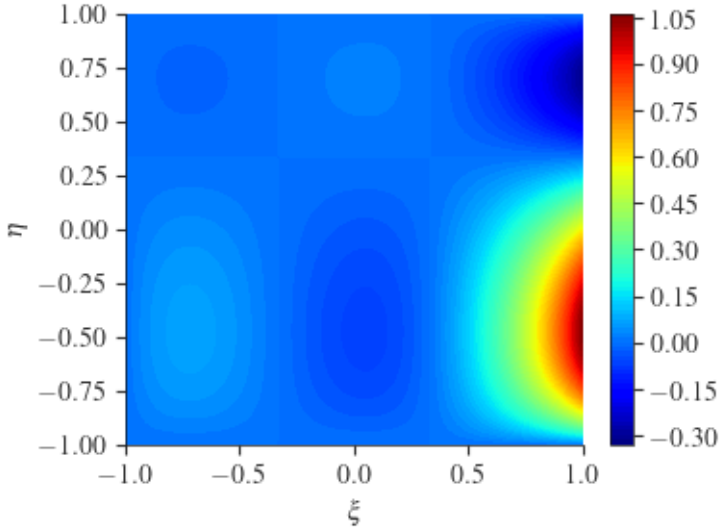
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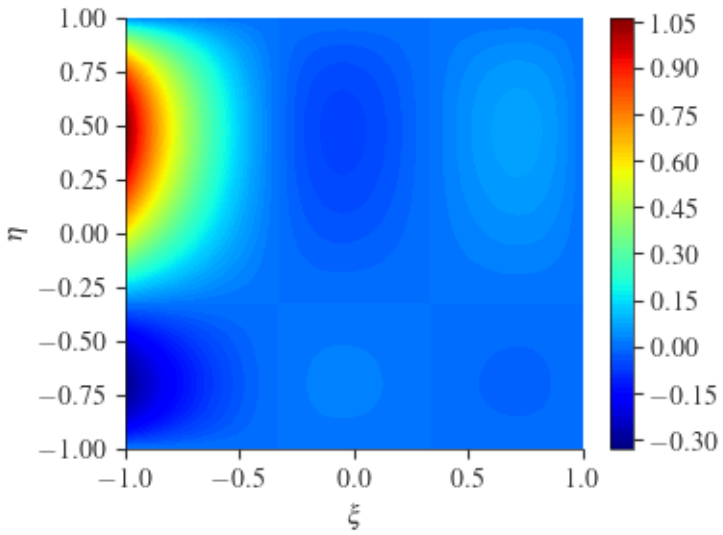
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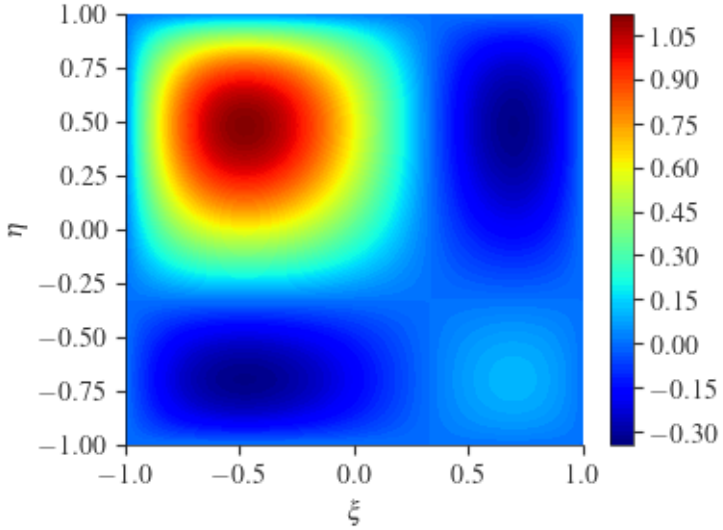
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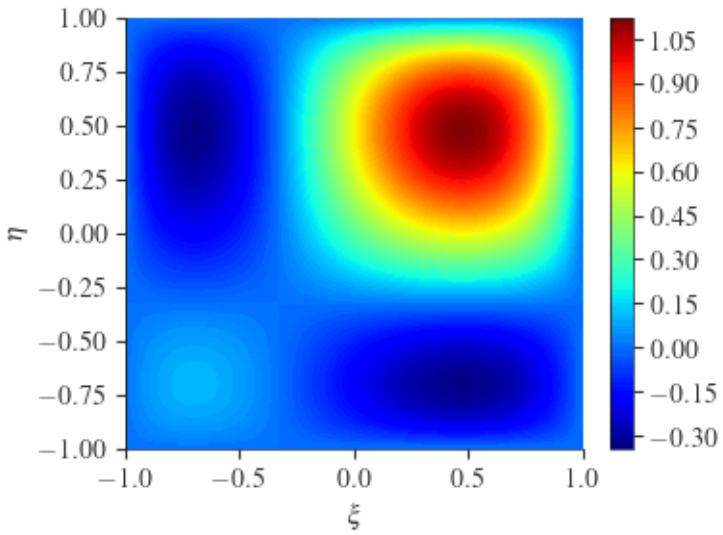
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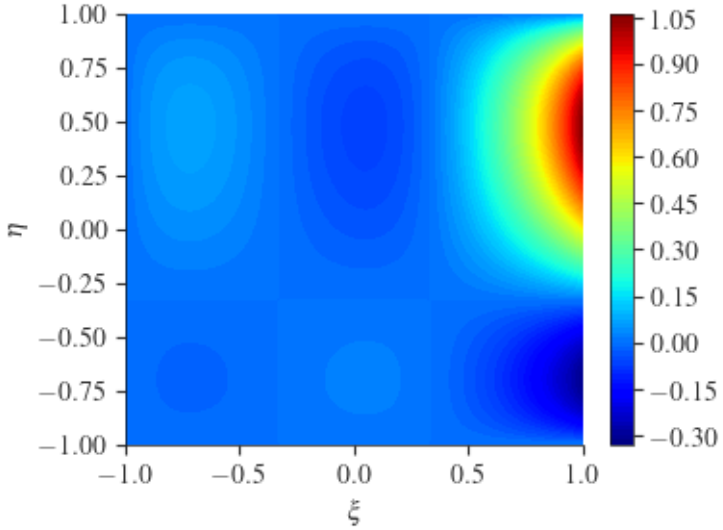
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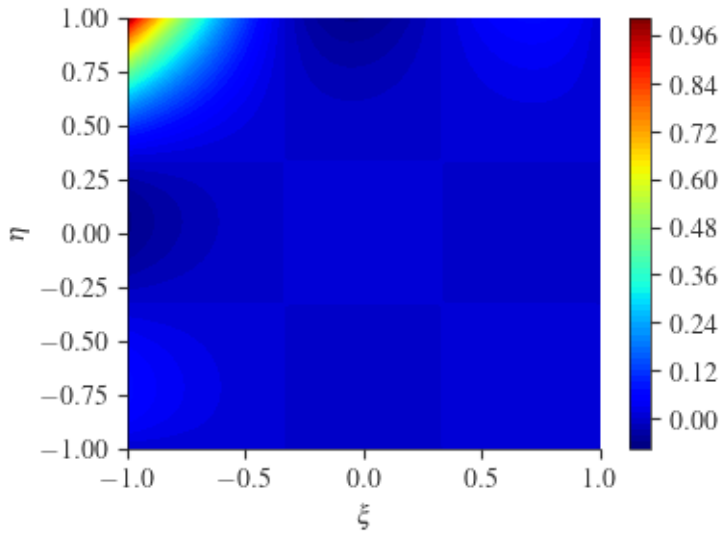
1.00
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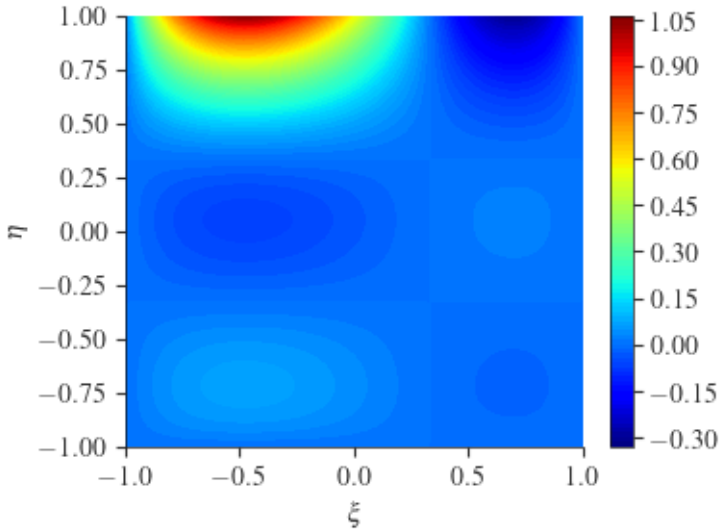
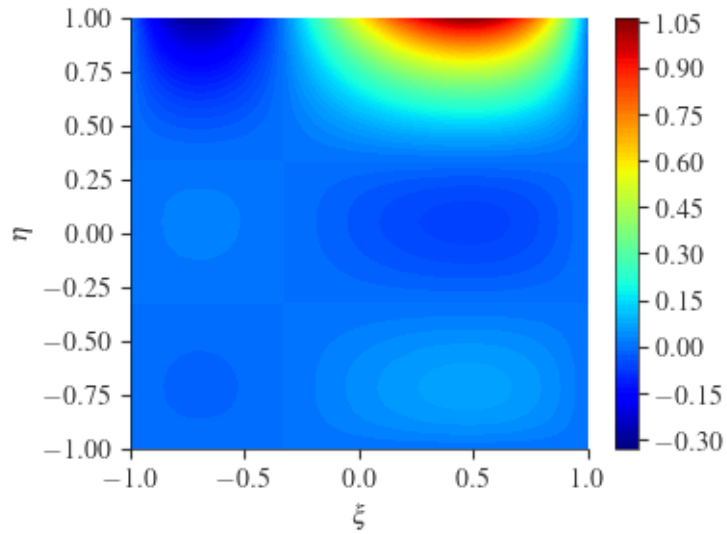
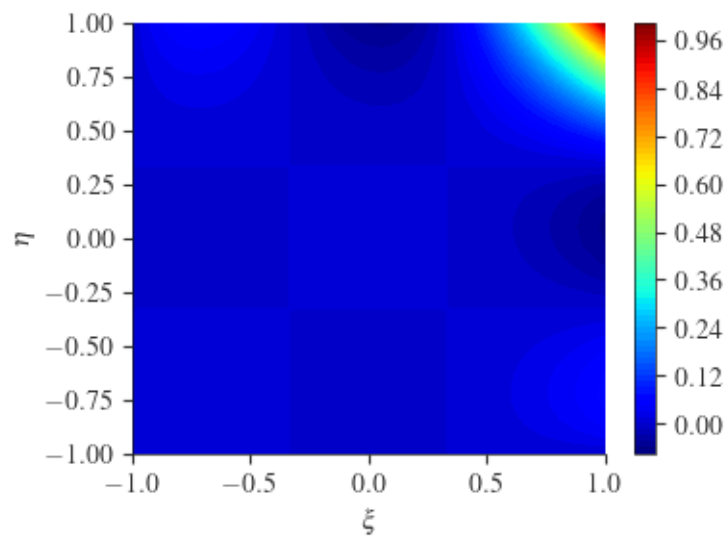
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.0
−0.5
0.0
0.5
1.0
ξ
0.96
0.84
0.72
0.60
0.48
0.36
0.24
0.12
0.00


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
η
−1.0
−0.5
0.0
0.5
1.0
ξ
1.05
0.90
0.75
0.60
0.45
0.30
0.15
0.00
−0.15
−0.30

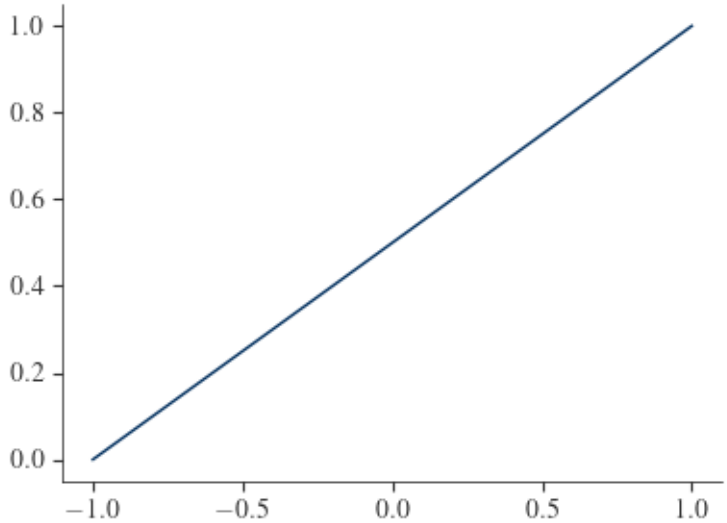




1.2 Problem 2

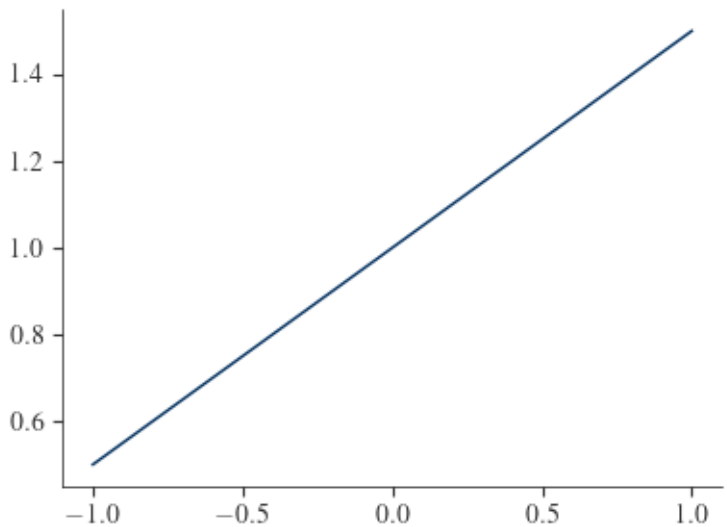
1.2.1 p=1, x_a={0,1}, ξ={-1,1}

[20]:
```
import CE_ME_507_UniDimensionalXMap as xmf
xmf.PlotXMap(1,[0.0,1.0],[-1.0,1.0],contours=False)
```



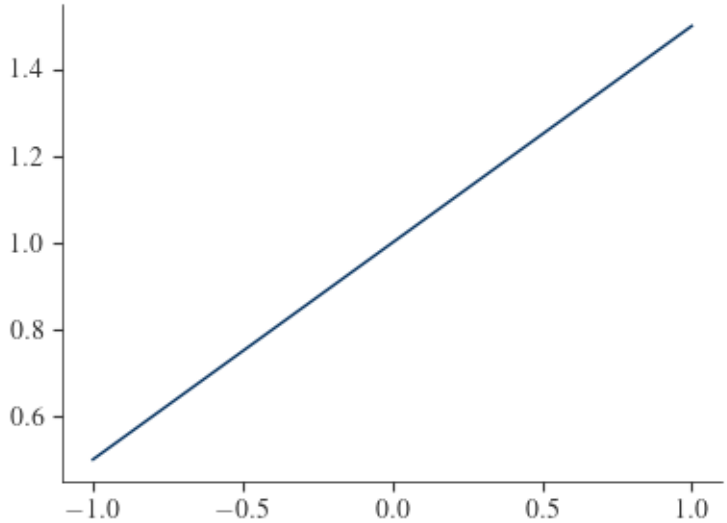
1.2.2 p=2, x_a={0.5,1,1.5}, ξ={-1,0,1}

[21]:
```
xmf.PlotXMap(2,[0.5,1.0,1.5],[-1.0,0.0,1.0],contours=False)
```



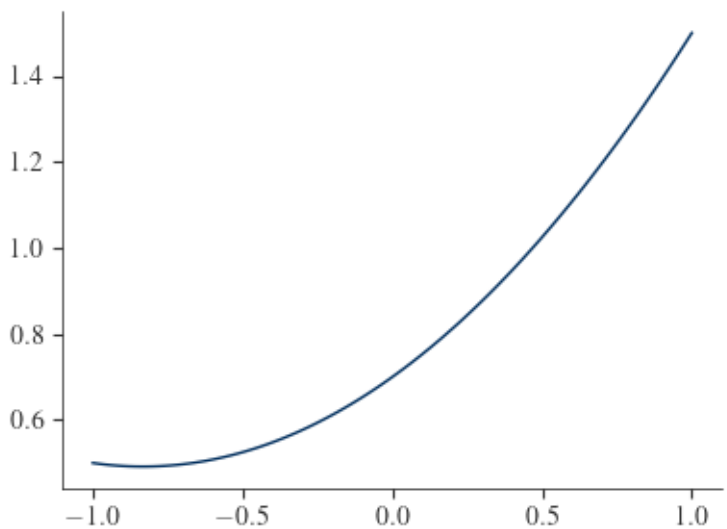
1.2.3 p=2, x_a={0.5,0.7,1.5}, ξ={-1,-0.6,1}

[22]:
```
xmf.PlotXMap(2,[0.5,0.7,1.5],[-1.0,-0.6,1.0],contours=False)
```



1.2.4 p=2, x_a={0.5,0.7,1.5}, ξ={-1,-0,1}

[23]:
```
xmf.PlotXMap(2,[0.5,0.7,1.5],[-1.0,-0,1.0],contours=False)
```



1.2.5 Questions

Question: When is the mapping $x^e(\xi)$ linear?
Answer: It is linear when the spacing of the non-end values of x_a and ξ_a are spaced the same fractional distance between the end points.

Question: When is this map transition from teh interval [-1,1] to the input sparital element in a manner that correcely transforms points for gaussian quadrature?
Answer: Only when the sampling distance is the same between all points.